Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			• {Peaks/maximum values} on absorption spectrum are at similar wavelengths to peaks on action spectrum/ At wavelengths (of light) where absorption {is higher/ increases} the rate of photosynthesis {is higher/ increases}/ora/ owtte (1) • Wavelengths (most) absorbed are the wavelengths used in photosynthesis. (1) Allow correct reference to values		1	1	2		2
	(b)	(i)		**Any one (x1) from** • Chlorophyll b has a peak {at approx. 640/ in the red (wavelengths)} **but** chlorophyll c does not (1) • Chlorophyll b absorbs longer wavelengths of light (1) • Chlorophyll b absorption increases around approx. 600 **but** chlorophyll c does not (1) • Chlorophyll c has one peak whereas Chlorophyll b has two (1) • Chlorophyll c absorbs less {at 640nm/ in the red (wavelengths)}/ ORA (1)		1		1		1
		(ii)		**Any one (x1) from** • {No peak/ lower rate of photosynthesis} at 640nm (1) • Narrower peak {at 670nm/ in red region} (1)			1	1		1
	(c)			**Any one (x1) from** • {Orange light/light of wavelength approx. 640} {does not penetrate water column/ does not reach 200m (use of figures)} (1) • (Advantage to have chlorophyll c) as blue light penetrates {further / to 300m / 200m} (use of figures) (1)			1	1		
	(d)			Domain-Eukaryota/e (1) Reason: has {nucleus/chloroplast/membranous/ membrane bound organelles} (1)		2		2		

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					AO1	AO2	AO3	Total	Maths	Prac
	(e)	(i)		44 $gm^{-3}day^{-1}$(1)		1		1	2	
		(ii)		$3.2x10^5$ = 3 marks **Award 2 marks for any of** 320000 8000000/25 8000/0.025 3.2×10^2 (not converted) **Award 1 mark for any of** 8000kg converted to g = 8000000 25g converted to kg = 0.025 320 (not converted and not standard form)		3		3	3	
				Question 2 total	**0**	**8**	**3**	**11**	**5**	**4**